MIT Assignment-07

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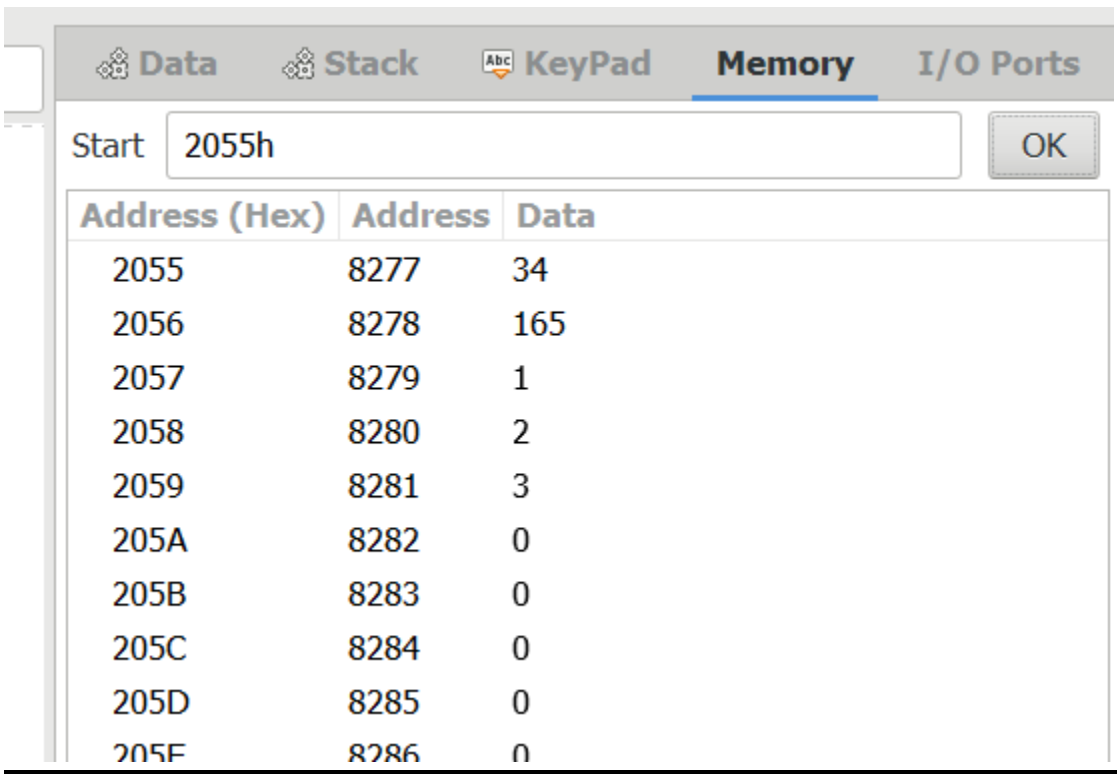
Question 1 : **the following block of data is stored in memory locations from 2055 to 205A.**

Transfer the data to the locations2080 to 2085 in the reverse order

Data(H) : 22,A5,01,02,03

Output:



Address (Hex)	Address	Data
2055	8277	34
2056	8278	165
2057	8279	1
2058	8280	2
2059	8281	3
205A	8282	0
205B	8283	0
205C	8284	0
205D	8285	0
205E	8286	0

Registers

A	22	
BC	00	00
DE	20	85
HL	20	54
PSW	00	00
PC	42	26
SP	FF	FF
Int-Reg	00	

Flag

S	0
Z	1
AC	0
P	1
C	0

Decimal - Hex Convertion

Decimal	Hex
0	0

To Hex | To Dec

I/O Ports

0 − + 00

Update Port Value

Memory

0 − + 0

Update Memory

Load me at

```
;code
start: nop
; ------------------------------
; Store data at 2055H - 2059H
; ------------------------------

LXI H, 2055H

MVI M, 22H
INX H

MVI M, 0A5H
INX H

MVI M, 01H
INX H

MVI M, 02H
INX H

MVI M, 03H


; ------------------------------
; Reverse transfer to 2080H-2084H
; ------------------------------

LXI H, 2059H
LXI D, 2080H
MVI C, 05H

LOOP: MOV A, M
      STAX D

      DCX H
      INX D

      DCR C
      JNZ LOOP

HLT

hlt
```

Data | Stack | KeyPad | Memory | I/O Ports

Start 2080h OK

Address (Hex)	Address	Data
2080	8320	3
2081	8321	2
2082	8322	1
2083	8323	165
2084	8324	34
2085	8325	0
2086	8326	0
2087	8327	0

Question 2:

2) A string of six data bytes is stored starting from memory location 2050H. The string includes some blanks (byte with zero value). Write a program to eliminate the blanks from the string.

Data (H) : F2, 08, 00, 40, 01, 00, A7

Output:

Registers

A	00	
BC	00	00
DE	20	54
HL	20	56
PSW	00	00
PC	42	2E
SP	FF	FF
Int-Reg	00	

Flag

S	0
Z	1
AC	0
P	1
C	0

Decimal - Hex Convertion

Decimal	Hex
0	0

To Hex | To Dec

I/O Ports

0 − + 00

Update Port Value

Memory

0 − + 00

Update Memory

Load me at

```
;code
start: nop

; Store data at 2050H

LXI H, 2050H

MVI M, 0F2H
INX H
MVI M, 08H
INX H
MVI M, 00H
INX H
MVI M, 40H
INX H
MVI M, 01H
INX H
MVI M, 00H


; ------------------------------
; Remove blanks (00 values)
; ------------------------------

LXI H, 2050H      ; HL → source
LXI D, 2050H      ; DE → destination (same place)
MVI C, 06H        ; Counter = 6

LOOP: MOV A, M    ; Load data

      CPI 00H     ; Check if zero
      JZ SKIP     ; If zero → skip

      STAX D      ; Store valid data
      INX D       ; Move destination

SKIP: INX H       ; Move source
      DCR C
      JNZ LOOP

HLT
```

Simulator: Idle

Data Stack KeyPad **Memory** I/O Po

Start 2050h

Address (Hex)	Address	Data
2050	8272	242
2051	8273	8
2052	8274	64
2053	8275	1
2054	8276	1
2055	8277	0

Question 3:

3) Write a program to add the following five data bytes stored in memory locations starting from XX60H & display the sum.

Data (H) : 1A, 32, 4F, 12, 27

Output:

Registers

A	E0	
BC	00	00
DE	00	00
HL	20	65
PSW	00	00
PC	42	26
SP	FF	FF
Int-Reg	00	

Flag

S	0
Z	1
AC	0
P	1
C	0

Decimal - Hex Convertion

Decimal	Hex
224	E0

To Hex | To Dec

I/O Ports

0 − + 00

Update Port Value

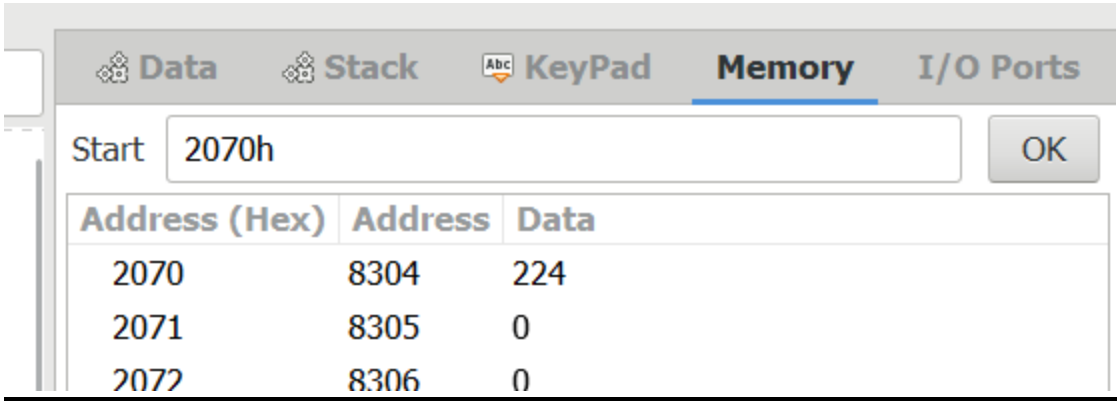
Memory

0 − + 00

Update Memory

Load me at

```
;<Program title>

jmp start

;data


;code
start: nop
; -------------------------------
; Store data at 2060H - 2064H
; -------------------------------

LXI H, 2060H
MVI M, 1AH
INX H
MVI M, 32H
INX H
MVI M, 4FH
INX H
MVI M, 1EH
INX H
MVI M, 27H


; -------------------------------
; Add 5 numbers
; -------------------------------

LXI H, 2060H      ; HL → start
MVI C, 05H        ; Counter = 5

MVI A, 00H        ; Clear accumulator

LOOP: ADD M       ; A = A + M
      INX H
      DCR C
      JNZ LOOP

STA 2070H         ; Store result at 2070H

HLT

hlt
```



Question 4 :

4) Write a program to count Number of times 05H appears in block of 10 numbers stored at memory location starts from 2000H and display count.

Data (H): 02, 05, 33, 05, 40, 50, 60, 80, 05, 05.

Output:

Registers

A	04	
BC	04	00
DE	00	00
HL	20	0A
PSW	00	00
PC	42	3C
SP	FF	FF
Int-Reg	00	

Flag

S	0
Z	1
AC	0
P	1
C	0

Decimal - Hex Convertion

Decimal	Hex
224	E0

To Hex | To Dec

I/O Ports

0 − + 00

Update Port Value

Memory

0 − + 00

Update Memory

Load me at

```
;code
start: nop
; ------------------------------
; Store data at 2000H - 2009H
; ------------------------------
LXI H, 2000H
MVI M, 02H
INX H
MVI M, 05H
INX H
MVI M, 33H
INX H
MVI M, 05H
INX H
MVI M, 40H
INX H
MVI M, 50H
INX H
MVI M, 60H
INX H
MVI M, 80H
INX H
MVI M, 05H
INX H
MVI M, 05H
; ------------------------------
; Count occurrences of 05H
; ------------------------------
LXI H, 2000H      ; HL → start
MVI C, 0AH        ; Counter = 10
MVI B, 00H        ; B = count
LOOP: MOV A, M    ; Load data
      CPI 05H     ; Compare with 05H
      JNZ SKIP    ; If not equal, skip
      INR B       ; Increment count

SKIP: INX H       ; Next memory
      DCR C
      JNZ LOOP

MOV A, B
STA 2010H         ; Store result
```

Simulator: Idle

Start 2010h OK

Address (Hex)	Address	Data
2010	8208	4
2011	8209	0

Question 5 :

5) A set of eight data bytes is stored in the memory location starting at XX50H. Check each data byte for bit D7 and D0. if D7 or D0 is 1, Reject the data byte. Otherwise store databytes at memory locations starting from XX60H.

Data (H) : 80, 52, E8, 78, F2, 67, 35, 62.

Output:

Registers

A	62	
BC	00	00
DE	20	63
HL	20	58
PSW	00	00
PC	42	3B
SP	FF	FF
Int-Reg	00	

Flag

S	0
Z	1
AC	0
P	1
C	0

Decimal - Hex Convertion

Decimal	Hex
224	E0

To Hex | To Dec

I/O Ports

0 − + 00

Update Port Value

Memory

0 − + 00

Update Memory

Load me at

```
start: nop
; ------------------------------
; Store data at 2050H - 2057H
; ------------------------------
LXI H, 2050H
MVI M, 80H
INX H
MVI M, 52H
INX H
MVI M, 0E8H
INX H
MVI M, 78H
INX H
MVI M, 0F2H
INX H
MVI M, 67H
INX H
MVI M, 35H
INX H
MVI M, 62H
; ------------------------------
; Filter data (D7=0 AND D0=0)
; ------------------------------
LXI H, 2050H      ; Source
LXI D, 2060H      ; Destination
MVI C, 08H        ; Counter = 8

LOOP: MOV A, M    ; Load data

      ANI 80H     ; Check D7
      JNZ SKIP    ; If D7=1 → reject

      MOV A, M    ; Reload original
      ANI 01H     ; Check D0
      JNZ SKIP    ; If D0=1 → reject

      MOV A, M    ; Reload original
      STAX D      ; Store valid data
      INX D

SKIP: INX H
      DCR C
      JNZ LOOP
HLT
```

Simulator: Idle

Question 6 :

Specify the contents of memory locations XX70H to XX74H after execution of the following instructions.

```
        LXI H,XX70H     ;Set up HL as a memory pointer
        MVI B,05H       ;Set up register B as a counter
        MVI A,01
STORE:  MOV M,A         ;Store (A) in memory
        INR A
        INX H
        DCR B
        JNZ STORE
        HLT
```

```
LXI H, XX70H
MVI B, 05H
MVI A, 01

STORE: MOV M, A
       INR A
       INX H
       DCR B
       JNZ STORE
HLT
```

Output:

Address	Data
XX70	01
XX71	02
XX72	03
XX73	04
XX74	05

Question 7 :

Identify the contents of the registers, the memory location (XX55H), and the flags as the following instructions are executed.

	A	H	L	S	Z	CY	M XX55H
LXI H,XX55H							
MVI M,8AH							
MVI A,76H							
ADD M							
STA XX55H							
HLT							

```
  76
+ 8A
-----
 100 (Hex)
```

Output:

Register / Flag	Value
A	00H
H	XX
L	55H
Memory (XX55H)	00H
S	0
Z	1
CY	1